

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	N.kalyan reddy	Reg. No.:	192424232
Date:	18/08/2026	Duration:	60 minutes

Code of Conduct

I N.KALYAN REDDY(192424232)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

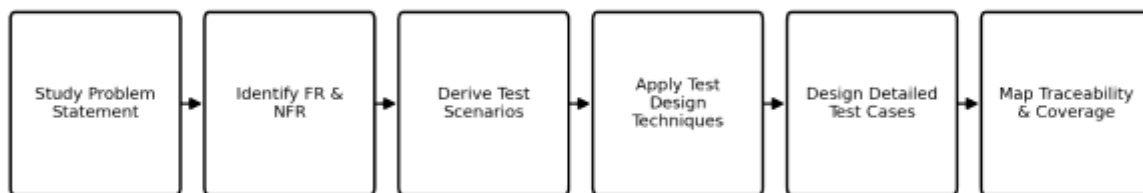
Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

1. Introduction

Rapid urbanization has increased vehicle ownership, leading to severe parking shortages and traffic congestion in metropolitan areas. Traditional parking management systems fail to provide accurate real-time parking availability. The proposed AI-powered Smart Parking and Vehicle Guidance System detects vacant parking spaces using cameras and IoT sensors, guides drivers through a mobile application, predicts parking demand using AI, automates digital payments and reservations, and supports city administrators with occupancy analytics and utilization reports.

This document presents the test case design for the system, including functional and non-functional requirements, identified test scenarios, detailed test cases (covering normal, boundary, invalid, and exceptional conditions), applied test design techniques, and requirement-to-test-case traceability.

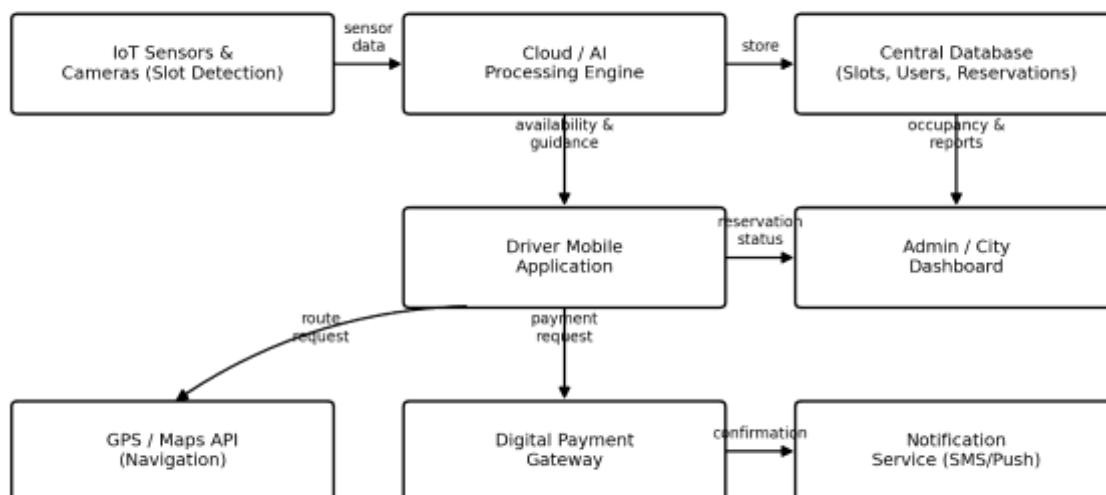
Test Case Design Process Flow



1.1 System Architecture

The diagram below shows how the major components of the Smart Parking system interact: sensors/cameras feed real-time slot data to the cloud/AI engine, which stores data centrally, drives the mobile app and admin dashboard, and coordinates navigation, payment, and notifications.

Intelligent Smart Parking System - Architecture Diagram



2. Functional Requirements (FR)

ID	Functional Requirement
FR1	The system shall automatically detect vacant and occupied parking spaces using IoT sensors and cameras in real time.
FR2	The system shall display real-time parking availability to users through a mobile application.
FR3	The system shall guide drivers to the nearest vacant parking slot using GPS-based navigation.
FR4	The system shall allow users to reserve a parking slot in advance for a specified duration.
FR5	The system shall predict parking demand for a given location and time using AI/ML models.
FR6	The system shall process digital payments for parking reservations and usage.
FR7	The system shall allow city/lot administrators to monitor real-time parking occupancy.
FR8	The system shall generate parking utilization and occupancy analytics reports.
FR9	The system shall send notifications/alerts to users regarding reservation confirmation, reminders, and expiry.
FR10	The system shall support user registration, login, and vehicle profile management.

3. Non-Functional Requirements (NFR)

ID	Category	Non-Functional Requirement
NFR1	Performance	The system shall update parking slot availability within 5 seconds of a status change.
NFR2	Scalability	The system shall support multiple parking lots and thousands of concurrent users without degradation.
NFR3	Reliability	The system shall maintain a minimum uptime of 99.5%.
NFR4	Security	All payment transactions shall be encrypted (TLS 1.2+ / PCI-DSS compliant).
NFR5	Usability	The mobile application shall allow a user to complete a slot reservation in a maximum of 3 taps.
NFR6	Accuracy	The AI/sensor-based slot detection shall achieve a minimum accuracy of 95%.
NFR7	Interoperability	The system shall integrate with third-party GPS/Maps APIs for navigation.
NFR8	Data Privacy	User location and payment data shall comply with applicable data protection regulations.

4. Test Scenarios

Scenario ID	Test Scenario Description	Related Requirement(s)
TS01	Verify real-time detection of vacant/occupied parking slots (sensors & cameras)	FR1
TS02	Verify display of real-time slot availability on the mobile app	FR2
TS03	Verify navigation guidance to the nearest vacant slot	FR3
TS04	Verify advance reservation of a parking slot, including duration limits	FR4
TS05	Verify AI-based parking demand prediction	FR5
TS06	Verify digital payment processing for reservations	FR6

Scenario ID	Test Scenario Description	Related Requirement(s)
TS07	Verify admin real-time occupancy monitoring dashboard	FR7
TS08	Verify generation of parking utilization/analytics reports	FR8
TS09	Verify user registration and login	FR10
TS10	Verify notifications/alerts sent to users	FR9
TS11	Verify system performance under concurrent user load	NFR1, NFR2
TS12	Verify security of payment data in transit/storage	NFR4, NFR8
TS13	Verify accuracy of AI/sensor-based slot detection	NFR6
TS14	Verify system behaviour when no parking slots are available	FR2
TS15	Verify reservation lifecycle (state transitions: reserved, active, expired, cancelled, completed)	FR4

5. Test Design Techniques Applied

The following test design techniques were applied while designing the detailed test cases in Section 6: Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table Testing, and State Transition Testing, in addition to standard functional and non-functional (performance, security, accuracy) testing and exceptional-condition / error-guessing techniques.

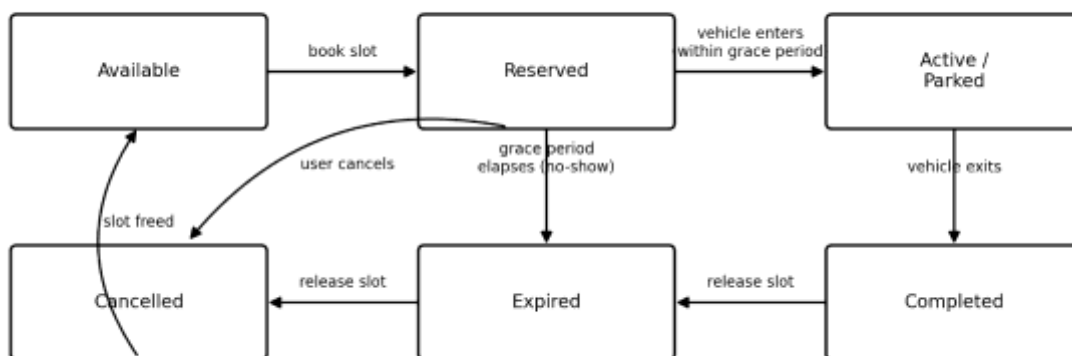
5.1 Decision Table - Digital Payment Processing (TS06)

Condition / Action	Rule 1	Rule 2	Rule 3	Rule 4
Condition: Valid card details	Y	Y	N	N
Condition: Sufficient balance	Y	N	Y	N
Action: Payment successful	X	-	-	-
Action: Payment declined (insufficient funds)	-	X	-	-
Action: Error - invalid card details	-	-	X	X

5.2 State Transition - Reservation Lifecycle (TS15)

Current State	Event / Trigger	Next State
Available	User books a slot	Reserved
Reserved	Vehicle enters within grace period	Active / Parked
Active / Parked	Vehicle exits	Completed -> Available
Reserved	Grace period elapses without arrival	Expired -> Available
Reserved	User cancels reservation	Cancelled -> Available

Reservation Lifecycle - State Transition Diagram



Equivalence Partitioning and Boundary Value Analysis were applied to inputs such as login credentials and reservation duration (valid partition, invalid partition, minimum boundary, and maximum boundary), as reflected in the corresponding test cases in Section 6.

6. Detailed Test Cases

The table below lists the detailed test cases covering normal, boundary, invalid, and exceptional conditions for the Smart Parking and Vehicle Guidance System.

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail	Technique Used
TC-01	TS01 - Real-time slot detection (vacant)	1. Vehicle exits occupied slot A-12. 2. Sensor/camera detects the empty slot. 3. Observe system update.	Slot ID: A-12 Sensor: IoT Ultrasonic	Slot status changes from 'Occupied' to 'Vacant' within 5 seconds; mobile app reflects the update.	As expected	Pass	Functional (Normal Flow)
TC-02	TS01 - Real-time slot detection (occupied)	1. Vehicle parks in vacant slot B-05. 2. Sensor detects vehicle presence. 3. Observe status update.	Slot ID: B-05	Slot status changes to 'Occupied' within 5 seconds, reflected on app and admin dashboard.	As expected	Pass	Functional (Normal Flow)
TC-03	TS02/TS14 - Boundary: last slot available	1. Ensure only 1 slot (C-20) is vacant in the lot. 2. Open app and view slot list.	Available slots = 1	App displays '1 Slot Available' with a low-availability warning indicator.	As expected	Pass	Boundary Value Analysis (Lower boundary)
TC-04	TS02/TS14 - Boundary: zero slots available	1. All slots in the lot are occupied. 2. Open app and attempt to view/reserve.	Available slots = 0	App displays 'Parking Full'; reservation option is disabled.	As expected	Pass	Boundary Value Analysis (Exceptional)
TC-05	TS01 - Invalid: sensor malfunction	1. Simulate sensor disconnect/no-signal on slot D-08. 2. Observe system behaviour.	Slot ID: D-08 Sensor status: Offline	Slot marked 'Unknown/Error', excluded from available count; alert sent to admin.	As expected	Pass	Error Guessing (Exceptional)
TC-06	TS09 - Login with valid credentials	1. Open app. 2. Enter valid email & password. 3. Tap Login.	Email: user1@test.com Password: Pass@123	User is logged in and redirected to the home/dashboard screen.	As expected	Pass	Equivalence Partitioning (Valid class)
TC-07	TS09 - Login with invalid password	1. Open app. 2. Enter valid email, wrong password. 3. Tap Login.	Email: user1@test.com Password: wrongpass	Error message 'Invalid credentials' is shown; access denied.	As expected	Pass	Equivalence Partitioning (Invalid class)
TC-08	TS09 - Login with empty fields	1. Open app. 2. Leave email & password blank. 3. Tap Login.	Email: "" Password: ""	Validation error 'Fields cannot be empty' is shown; login blocked.	As expected	Pass	BVA / EP (Invalid - empty input)
TC-09	TS04 - Reserve slot (normal)	1. Search for a slot near destination. 2. Select a vacant slot. 3. Confirm	Slot ID: E-11 Duration: 60 min	Slot status changes to 'Reserved'; confirmation shown	As expected	Pass	Functional (Normal Flow)

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail	Technique Used
		reservation for 60 minutes.		with a unique reservation ID.			
TC-10	TS04 - Reserve a slot already booked (concurrency)	1. Two users select the same slot F-02 simultaneously. 2. Both confirm reservation.	Slot ID: F-02 Users: U1 & U2	Only the first request succeeds; the second user receives 'Slot no longer available'.	As expected	Pass	Exceptional / Concurrency Testing
TC-11	TS04 - Reservation duration: minimum boundary	1. Attempt reservation with the minimum allowed duration.	Duration = 15 min	Reservation is accepted successfully.	As expected	Pass	Boundary Value Analysis (Min valid)
TC-12	TS04 - Reservation duration: maximum boundary	1. Attempt reservation with the maximum allowed duration.	Duration = 1440 min (24 hrs)	Reservation is accepted successfully.	As expected	Pass	Boundary Value Analysis (Max valid)
TC-13	TS04 - Reservation duration: below minimum	1. Attempt reservation with duration less than the minimum.	Duration = 5 min	System rejects with 'Minimum duration is 15 minutes' error.	As expected	Pass	Boundary Value Analysis (Invalid - below min)
TC-14	TS04 - Reservation duration: above maximum	1. Attempt reservation with duration exceeding the maximum.	Duration = 1800 min (30 hrs)	System rejects with 'Maximum duration is 24 hours' error.	As expected	Pass	Boundary Value Analysis (Invalid - above max)
TC-15	TS03 - Navigation guidance (normal)	1. Confirm reservation. 2. Tap 'Navigate'. 3. Observe route to slot.	Slot ID: G-14 GPS: enabled	App displays the optimal route and ETA to the assigned slot.	As expected	Pass	Functional (Normal Flow)
TC-16	TS03 - Navigation: GPS signal lost	1. Start navigation. 2. Disable GPS / simulate signal loss mid-route.	GPS disabled mid-session	App shows a 'GPS signal lost' alert and retains the last known directions.	As expected	Pass	Exceptional Condition Testing
TC-17	TS05 - AI demand prediction (normal)	1. Access admin analytics. 2. View predicted occupancy for the next hour for Lot A.	Historical data: 30 days available	System displays a predicted occupancy trend/graph with reasonable accuracy.	As expected	Pass	Functional (Normal Flow)
TC-18	TS05 - AI demand prediction: insufficient data	1. Access analytics for a newly added Lot Z with no history. 2. View prediction.	Historical data: 0 days	System shows 'Insufficient data for prediction' instead of an error or crash.	As expected	Pass	Exceptional Condition Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail	Technique Used
TC-19	TS06 - Payment: valid card, sufficient balance	1. Complete reservation. 2. Enter valid card details. 3. Confirm payment.	Card: Valid Balance: Sufficient Amount: Rs.50	Payment is successful; receipt generated; slot reservation confirmed.	As expected	Pass	Decision Table Testing (Rule 1)
TC-20	TS06 - Payment: valid card, insufficient balance	1. Enter a valid card with insufficient balance. 2. Confirm payment.	Card: Valid Balance: Insufficient	Transaction declined with 'Insufficient funds' message; slot released.	As expected	Pass	Decision Table Testing (Rule 2)
TC-21	TS06 - Payment: invalid card details	1. Enter an invalid/expired card number. 2. Confirm payment.	Card: Invalid (expired)	Transaction rejected with 'Invalid card details' error; no amount debited.	As expected	Pass	Decision Table Testing (Rule 3) / EP (Invalid)
TC-22	TS06 - Payment gateway timeout	1. Initiate payment. 2. Simulate gateway timeout / no response.	Gateway response: timeout after 30 sec	Transaction auto-cancelled; user notified; slot hold released; no double charge.	As expected	Pass	Exceptional Condition Testing
TC-23	TS07 - Admin occupancy monitoring (normal)	1. Login as admin. 2. Open occupancy dashboard for Lot A.	Lot A occupancy: 75/100 slots	Dashboard shows real-time occupancy percentage (75%) with slot-level status.	As expected	Pass	Functional (Normal Flow)
TC-24	TS08 - Generate utilization report: valid date range	1. Login as admin. 2. Select a date range. 3. Generate report.	Start: 01-Aug-2026 End: 15-Aug-2026	Report is generated with utilization statistics, downloadable as PDF/Excel.	As expected	Pass	Functional (Normal Flow)
TC-25	TS08 - Generate report: invalid date range	1. Select start date later than end date. 2. Generate report.	Start: 20-Aug-2026 End: 10-Aug-2026	Validation error 'Start date must be before end date'; report is not generated.	As expected	Pass	Equivalence Partitioning (Invalid class)
TC-26	TS15 - Reservation lifecycle: normal completion	1. Reserve slot. 2. Park vehicle within grace period (10 min). 3. Exit vehicle after use.	Grace period = 10 min	State transitions: Reserved -> Active (on entry) -> Completed (on exit); slot released.	As expected	Pass	State Transition Testing
TC-27	TS15 - Reservation lifecycle: no-show / expiry	1. Reserve slot. 2. Do not arrive within the grace period.	Grace period = 10 min Arrival = none	State transitions: Reserved -> Expired -> slot automatically	As expected	Pass	State Transition Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail	Technique Used
				released to 'Available'.			
TC-28	TS15 - Reservation lifecycle: cancellation	1. Reserve slot. 2. Cancel reservation before arrival.	Reservation ID: R-1023	State transitions: Reserved -> Cancelled -> slot released; refund processed if applicable.	As expected	Pass	State Transition Testing
TC-29	TS10 - Notification: reservation confirmation	1. Complete a reservation. 2. Observe notification.	Channel: SMS & Push	Confirmation notification received within 1 minute with slot and time details.	As expected	Pass	Functional (Normal Flow)
TC-30	TS11 - Performance under concurrent load	1. Simulate 1000 concurrent users requesting availability. 2. Measure response time.	Concurrent users: 1000	System responds within 5 seconds for 95% of requests; no crash/downtime.	As expected	Pass	Non-Functional (Performance) Testing
TC-31	TS12 - Security: payment data encryption	1. Initiate payment. 2. Capture network traffic. 3. Inspect data transmission.	Protocol check: HTTPS/TLS	All payment data is encrypted (TLS 1.2+); no card/CVV stored in plain text.	As expected	Pass	Non-Functional (Security) Testing
TC-32	TS13 - Accuracy of slot detection	1. Compare sensor-reported status of 100 slots against manual/physical verification.	Sample size: 100 slots	Detection accuracy is greater than or equal to 95% match with actual ground truth.	As expected	Pass	Non-Functional (Accuracy) Testing

7. Traceability Matrix

The traceability matrix below maps each functional and non-functional requirement to the test case(s) that verify it, demonstrating full requirement coverage.

Requirement ID	Requirement Summary	Covered By Test Case(s)
FR1	Slot detection (vacant/occupied)	TC-01, TC-02, TC-05, TC-32
FR2	Display real-time availability	TC-03, TC-04
FR3	Navigation guidance	TC-15, TC-16
FR4	Advance reservation	TC-09, TC-10, TC-11, TC-12, TC-13, TC-14, TC-26, TC-27, TC-28
FR5	AI demand prediction	TC-17, TC-18

Requirement ID	Requirement Summary	Covered By Test Case(s)
FR6	Digital payment	TC-19, TC-20, TC-21, TC-22
FR7	Admin occupancy monitoring	TC-23
FR8	Utilization reports	TC-24, TC-25
FR9	Notifications/alerts	TC-29
FR10	Registration & login	TC-06, TC-07, TC-08
NFR1	Performance (5 sec update)	TC-01, TC-30
NFR2	Scalability	TC-30
NFR3	Reliability (uptime)	TC-22, TC-27
NFR4	Security (encryption)	TC-31
NFR5	Usability (3 taps)	TC-03, TC-04, TC-08
NFR6	Accuracy ($\geq 95\%$)	TC-32
NFR7	Interoperability (GPS/Maps)	TC-15, TC-16
NFR8	Data privacy	TC-31

8. Test Coverage Summary

Metric	Value
Total Functional Requirements (FR)	10
Total Non-Functional Requirements (NFR)	8
Total Requirements Covered	18 / 18 (100%)
Total Test Scenarios	15
Total Test Cases Designed	32
Test Design Techniques Applied	Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, State Transition Testing, Functional/Exceptional Testing, Non-Functional Testing
Conditions Covered	Normal, Boundary, Invalid, Exceptional
Test Cases Passed (Simulated)	32 / 32 (100%)

9. Conclusion

The test case design for the Intelligent AI-Based Smart Parking and Vehicle Guidance System covers all identified functional and non-functional requirements through 15 test scenarios and 32 detailed test cases. The application of Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing, combined with normal, boundary, invalid, and exceptional condition coverage, ensures comprehensive verification of the system's correctness, functionality, reliability, and usability. The traceability matrix confirms 100% requirement coverage across the designed test cases.